

Supplement: Customer Base Change and Approved Water Charges in Kentucky 2019 to 2024

October 2, 2026. The accompanying CSV files preserve district identifiers, source references, and full numerical precision. Inputs are in data/ and generated results in results/. This supplement gives the complete sample disposition, diagnostics, and documentary qualifications.

1 Complete sample disposition

Table S1 reports the price and account decisions for every selected district. “Decline” and “Nondecline” are original sampling strata, assigned before tariff reconstruction. “Supported” and “Conditional” both indicate a reconstructed endpoint pair; “No pair” means none accepted. “Pass” means passing the stated six-year reporting screen; it does not certify stable physical boundaries or perfect category continuity. Price support is assessed separately. The account screen uses annual end-count changes above 10%, opening/prior-closing discrepancies above 1%, report completeness, reported county sets and documented mergers. These thresholds are pragmatic, not externally validated optimal cutoffs.

Table S1. Complete disposition of the 34 selected districts

District and ID	Original stratum	Price pair	Account screen	Primary	Decision and evidence
Barkley Lake (18500)	Nondecline	Supported	Fail	No	Residential/unmetered discontinuity, county-set change and count jumps unresolved. Source: District decision memo: Barkley; reports 2021/2023 p49 and county p12.
Bullock Pen (19200)	Nondecline	No pair	Pass	No	No accepted endpoint pair in retained reconstruction; extension-charge applicability unresolved. Geographic exceptions alone are not a categorical ban. Source: Original tariff feasibility/disposition; no completed non-extension pair.
Caldwell Co. (19201)	Decline	Supported	Fail	No	Residential/unmetered classification and 2023–24 opening gap unresolved. Source: District decision memo: Caldwell; 2022/2024 reports p49.
Corinth (19900)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.

District and ID	Original stratum	Price pair	Account screen	Primary	Decision and evidence
Cumberland Falls Hwy. (20200)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
East Pendleton (21200)	Nondecline	Supported	Fail	No	Annual increase exceeds 10%; financial discussion of new customers does not reconcile net accounts. Source: District decision memo: East Pendleton; reports p49, financial statement p10.
Estill Co. #1 (21500)	Nondecline	Supported	Pass	Yes	Retain non-Cobhill benchmark paired with district-wide counts; area shares and sub-threshold swing unresolved. Source: Comparability memo: Estill; 2023-00371.
Garrison (21900)	Nondecline	Conditional	Fail	No	Non-extension benchmark possible, but \$1.73 endpoint charge remains conditional; separate account opening gap fails screen. Source: Comparability memo: Garrison; current tariff pp5–6; 2023-00408.
Hardin Co. #1 (22500)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Henderson Co. (22700)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Henry Co. #2 (23000)	Decline	Supported	Fail	No	2023 account report unavailable. Corrected 2024 tariff retained in broader sample. Source: District decision memo: Henry; case 2023-00090 Appendix A.
Hyden-Leslie Co. (23300)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.

District and ID	Original stratum	Price pair	Account screen	Primary	Decision and evidence
Knott Co. (19400)	Nondecline	Supported	Pass	Yes	Admitted in v3: southwest usage table cancelled in 2007; ordinary category traceable; unchanged account screen passes. Source: Comparability memo: Knott; tariff pp1/5/6; 2019-00268.
Laurel Co. #2 (24100)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Letcher Co. (7000300)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Lyon Co. (24500)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Martin Co. (25000)	Decline	Conditional	Pass	No	Debt-surcharge payoff/cap status unresolved at final endpoint; listed-charge scenario only. Source: V3 comparability memo; original tariff disposition, Martin charge evidence.
Mountain (25605)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Muhlenberg Co. #3 (26000)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Nebo (26400)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.

District and ID	Original stratum	Price pair	Account screen	Primary	Decision and evidence
North Hopkins (26600)	Decline	Supported	Fail	No	Temporary 1,376 → 1,598 → 1,365 residential sequence unexplained. Source: District decision memo: North Hopkins; 2022/2023 reports p50.
North McLean Co. (26900)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
North Mercer (27000)	Decline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Pendleton Co. (28000)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
Powell's Valley (28300)	Decline	Supported	Pass	Yes	Retain ordinary benchmark: order/reports support three counties; first surcharge invoice remains unknown. Source: Comparability memo: Powell; 2023-00387 p5.
Rattlesnake Ridge (28600)	Decline	Conditional	Fail	No	Conditional surcharge applicability and start/prior-end discrepancy. Source: Original price disposition; selected-34 screen; Rattlesnake charge notes.
Sandy Hook (29200)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.
South Anderson (29800)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.

District and ID	Original stratum	Price pair	Account screen	Primary	Decision and evidence
Southern Madison (30600)	Nondecline	Conditional	Fail	No	Conditional fee/surcharge treatment and large account-category/continuity flags. Source: Original price disposition; selected-34 screen; Southern Madison charge notes.
Southern Water & Sewer (7000900)	Decline	No pair	Fail	No	Meter-installation transition and reported county-set change prevent a supported pair. Source: Original tariff disposition; selected-34 screen; Southern Water & Sewer.
Trimble Co. #1 (31200)	Decline	Supported	Fail	No	Residential fall coexists with commercial/total growth; exact classification crosswalk unavailable. Source: District decision memo: Trimble; 2021–2023 reports p49.
Union Co. (31400)	Nondecline	Conditional	Pass	No	Surcharge collection-cap status unresolved; endpoint scenario conditional. Source: V3 comparability memo; original tariff disposition, Union charge evidence.
Webster Co. (31800)	Decline	Supported	Fail	No	2019 closing versus 2020 opening discrepancy of 100 residential accounts. Source: District decision memo: Webster; 2019/2020 reports p50.
Western Fleming Co. (32500)	Nondecline	Supported	Pass	Yes	Supported benchmark; passes stated mechanical screen. Physical-boundary stability not certified. Source: Endpoint and comparability registers; original account screen.

The table reconciles to 34 selected districts, 32 reconstructed pairs, 27 supported pairs and 20 primary observations. Five reconstructed pairs are displayed as conditional; Garrison retains the original “regional” code in the underlying panel. Bullock Pen and Southern Water & Sewer lack accepted reconstructed pairs. Bullock Pen is not excluded by a universal rule against geographic supplements: a completed supported non-extension pair is unavailable in the retained work. If such evidence is obtained, its price eligibility must be reconsidered consistently.

Among the 27 supported pairs, the seven account exclusions are East Pendleton, Trimble, North Hopkins, Webster, Henry #2, Barkley and Caldwell. Their detailed decisions and original page references remain in docs/evidence.md and sources/catalog.csv. Retention is 20/34 = 58.8%, 9/17 = 52.9% among initial

decliners and $11/17 = 64.7\%$ among sampled nondecliners; these retention rates do not determine the direction of selection bias.

Knott remains included: cancelled southwest usage rates do not establish an alternative ordinary endpoint category. Estill remains an explicitly non-Cobhill benchmark, with whole-district accounts and unknown shares; its sub-threshold swing is unexplained. Powell's three-county service is supported by both reports and its 2024 order. Garrison's ordinary non-extension benchmark does not resolve the separate \$1.73 component ambiguity or account opening discrepancy. The full comparability decisions are preserved in docs/evidence.md.

2 Full-sample documentary synthesis

Manuscript Appendix B contains the readable arithmetic table and companion documentary table for all 20 primary districts, in the same order as Appendix A. results/Reconstruction_Summary_20_v4.csv supplies their full-precision data and references. Eighteen have higher endpoint base charges; two have unchanged benchmarks. Four have positive net endpoint extras. The sums \$207.14 base + \$7.33 extras = \$214.47 are unweighted hypothetical-charge identities, not utility revenue or causal shares. Hyden's \$1.53 temporary surcharge is absent at both endpoints. Estill's \$3.54 is present at both. In either case, a zero endpoint difference in extras does not mean that no surcharge was billed during the interval.

Proceeding labels describe regulatory routes; economic interpretations require actual documentary support. Unknown intermediate reasons remain unknown. Pendleton E057's generic v3 rate-review label is corrected to wholesale special-contract amendment, based on the October 28, 2020 order in 2020-00158, p1. The ordinary residential base stays unchanged. This is a documentary classification correction only.

3 Current exploratory diagnostics

Account change is $c = 100 \times (N_{2024}/N_{2019} - 1)$. For band b, decline is $c < -b$, growth is $c > b$, and neutral is $|c| \leq b$. Bands are ordinary percentage units, not proportions or log points. Neutral observations are excluded only from these median comparisons; primary regression membership and continuity screening are unchanged.

Table S2. Current neutral-band descriptions

Band %	Decline n	Neutral n	Growth n	Decline median %	Growth median %	Gap pp
0	9	0	11	34.6862	24.2431	10.4431
0.1	8	1	11	34.3084	24.2431	10.0654
1	7	2	11	33.9307	24.2431	9.6877

The current declining-versus-growing median growth gap is 10.44 percentage points. It incorporates Knott's inclusion after review of its cancelled tariff and Muhlenberg's corrected endpoint. The consequential source corrections are documented in docs/evidence.md. Group mean charges are

\$36.89 → \$49.77 for decline and \$42.70 → \$51.66 for growth, with mean dollar changes \$12.89 and \$8.95. Different percentage denominators do not establish convergence or delayed adjustment.

Table S3. Current geographic diagnostics

Sample	N	Slope	95% HC3 interval
All	20	-1.231092	[-2.793759, 0.331575]
Omit Estill	19	-1.196907	[-2.791146, 0.397332]
Omit Powell	19	-1.633053	[-3.139395, -0.126712]
Omit both	18	-1.600812	[-3.139345, -0.062278]

Omitting Powell, alone or with Estill, produces an interval below zero, but source evidence supports their designated benchmarks. These exploratory omissions are not eligibility tests. Leave-one-out slopes run from -1.633053 (omit Powell) to -0.885785 (omit Cumberland Falls Highway); Hyden has the largest Cook's distance, and its omission gives -0.921579. The incomplete supplier audit identifies South Anderson → North Mercer and Knott → Letcher, not a complete dependence network. HC3 does not account for supplier or regional dependence.

Table S4. Retained current specifications

Specification	N	Slope	95% HC3 interval
Primary	20	-1.231092	[-2.793759, 0.331575]
Unadjusted	20	-1.276098	[-2.622746, 0.070550]
Broader supported	27	-0.515251	[-1.250824, 0.220321]
Base only	20	-1.170207	[-2.782348, 0.441934]
Add log initial charge	20	-0.976681	[-2.335263, 0.381900]
Add raw initial charge	20	-0.920582	[-2.277673, 0.436509]
Original design weights	20	-1.598464	[-3.206145, 0.009217]
Dollar change	20	-0.570730	[-1.255517, 0.114057]
Omit Hyden-Leslie	19	-0.921579	[-2.262956, 0.419798]
2000 gallons	20	-1.013792	[-2.428136, 0.400552]
6000 gallons	20	-1.339816	[-2.994083, 0.314451]
Exclude mixed; ARC adjusted	19	-1.477797	[-3.410690, 0.455096]
Martin conditional scenario	21	-0.727159	[-2.210215, 0.755898]

Table S4 reports all 13 specifications, using the current data and eligibility rules. The dollar-change row has dollars as its outcome; other rows use 100 times a log charge ratio. No added regression was selected for significance. The raw-initial-charge adjustment and log-initial-charge adjustment are explicitly distinct. Original sampling weights do not repair later exclusions.

4 Documentary counting units

The ledger contains 68 schedule/authorization records, including 20 baseline anchors and 48 subsequent entries. Forty-five subsequent entries change the benchmark arithmetic and three do not. One is a superseded same-effective-date approval. Eight amendment/clarification and six recurring-charge records are coded separately: 82 logical records in total, carrying 43 distinct non-null case IDs. Twenty-four schedule rows lack a linked case ID. These are not counts of independent proceedings, implemented price changes or invoice periods. Legally operative period and distinct implemented-change totals remain unknown.

All 20 base-price paths reconcile arithmetically. That cannot exclude missing intermediate reversals. Three surcharge records support a first billing month (Hyden, Mountain and reported Henderson); Western Fleming supports a reported collection start only. Powell's first invoice and Estill's complete billing spell are unavailable. The detailed count reconciliation is preserved in docs/evidence.md. Its numerical counts remain current; the one proceeding-label correction is recorded above and in results/events_v4.json.

5 Financial definitions and source qualifications

Financial definitions and documentary interpretation

The financial definitions below support the manuscript's interpretation of approved rates. The existing 40-endpoint tariff coding and account decisions are retained. Targeted source inspection is distinct from independent double coding of every tariff.

Knott financial definitions

Source: January 31, 2020 PSC order, case 2019-00268, preserved as sources/files/knott_order.pdf. Original order (https://psc.ky.gov/pscscf/2019%20Cases/2019-00268/20200131_PSC_ORDER.pdf). The financial table is on PDF p35, the different summary amount on p37, and the approved phases on pp38 and 41–42. The accompanying expense and debt-service discussion is on pp32–34 and p43.

Quantity	Amount	Definition and page
Pro-forma operating expenses	\$2,908,014	Final adjusted expense amount, p35; Appendix B p43
Average annual principal and interest	\$47,791	2020–2024 debt-service average, pp33/35
Additional working capital	\$9,558	Added requirement, p35
Overall revenue requirement	\$2,965,363	Sum of preceding three items, pp35/37
Other operating revenue	\$33,930	Deduction in retail requirement calculation, p35
Interest income	\$975	Deduction, p35
Nonutility income	\$8,846	Deduction, p35
Proposed wholesale sales revenue	\$1,250,850	Deduction before retail residual, p35
Revenue required from rates	\$1,670,762	Retail residual after those deductions, p35
Pro-forma present-rate service revenues	\$992,944	Existing retail revenue baseline, p35; retail sales label confirmed p43
Required retail revenue increase	\$677,818	\$1,670,762 – \$992,944; 68.26% of retail baseline, p35
Summary water-service revenue increase	\$646,151	Summary assertion, p37; no matching calculation shown there

The p35 table reconciles internally: $2,908,014 + 47,791 + 9,558 = 2,965,363$; less $33,930 + 975 + 8,846 + 1,250,850$ leaves $1,670,762$; less $992,944$ gives $677,818$. The p37 amount is \$31,667 lower. The inspected pages do not establish an exact bridge to that summary amount. Appendix B uses \$1,148,640 in adjusted present wholesale revenue, while p35 deducts proposed wholesale revenue; those are different concepts, and their difference does not explain \$31,667. The original staff recommendation on p3 is \$698,176, also not \$646,151. No claim that the summary is a typo, or that the retail and water-service figures are interchangeable, is justified here.

The main paper uses the reconciled overall requirement and approved phases, and omits both competing revenue-increase amounts as numerical findings. They are preserved here with definitions. The benchmark calculations remain $\$29.51 \rightarrow \$43.24 \rightarrow \$49.80$; pp38, 41–42 establish phase-one service on the order date and phase two on its first anniversary. The order describes a two-year phase-in to reduce rate shock (p36). Revenue requirements are regulatory pro-forma constructs; neither required revenue nor standardized price growth is observed cost growth or receipts.

The current panel gives Knott accounts $3,055 \rightarrow 3,170$, +3.7643208%, and charge growth +68.7563538%. Hyden accounts $3,412 \rightarrow 3,372$, –1.1723329%, and charge growth +86.8170267%. This is a comparison of heterogeneous cases, not matched causal controls.

Institutional setting and record classification

The official PSC overview (<https://psc.ky.gov/Home/About>) supports jurisdiction over water districts and the distinction from municipal retail regulation. The PSC PWA FAQ (<https://psc.ky.gov/Home/PurchasedWaterAdjustment>) explains wholesale-cost pass-through and both upward and downward adjustments. Archived HTML, URLs, retrieval times and hashes are in [sources/catalog.csv](#).

Original orders give the study-specific detail: Corinth 2021-00425 pp1–2 describes alternative-rate review; Hardin 2022-00337 pp1–2 explains the financing-linked framework; Nebo 2024-00062 pp1–2 identifies a loan-required rate increase linked to a pump/SCADA/meter project. Laurel 2020-00079 and Western Fleming 2019-00456 p1 likewise link project/financing approvals and rates. These links do not identify a causal dollar share attributable to investment.

Pendleton's October 28, 2020 order, case 2020-00158 p1, concerns amendment of its wholesale special contract with the City of Butler. V3's generic rate-review label for E057 is corrected in [results/events_v4.json](#). The ordinary residential base is unchanged at that entry. No event is added or removed; the 68 schedule/authorization, eight amendment and six recurring-charge totals remain unchanged. The 43 distinct recorded case IDs also remain unchanged.

Unclassified intermediate steps (Lyon 2024, Muhlenberg 2022, North McLean February 2021, North Mercer 2021, Sandy Hook 2022) remain unclassified at the stated level rather than acquiring an inferred cause. The full-sample companion table distinguishes components, proceedings and document-supported interpretations.

Mountain surcharge cap

The September 2, 2026 order in case 2025-00327 contains a further internal discrepancy. The narrative on PDF p38 describes the existing \$1.72 surcharge as limited to 48 months or \$1,365,608 assessed. Appendix B, PDF p49, instead specifies four years from January 2024 or \$1,368,516 collected, in both phase schedules. The caps differ by \$2,908, and the assessed/collected wording also differs. Both duration formulations support 48 months, but neither expressly reconciles the April 2024 order's January 2024–December 2028 dates.

These statements are preserved without choosing a corrected cap or inferring a cessation date. They create uncertainty about the surcharge's cessation conditions and later billing history. They do not provide evidence that the supported December 2024 \$1.72 component, or the \$56.63 benchmark including it, should change. The event data continue to record the original authorization cap; neither amount is substituted for the other as a resolved cessation condition. Source: [sources/files/mountain_2026_order.pdf](#), pp38, 49; [sources/files/mountain_april_order.pdf](#), pp3–4.

6 Literature and evidence boundaries

The Pennsylvania predecessor is Bash, Grimshaw, Horan, Stanmyer, Warren and Patterson (2020), Addressing Financial Sustainability of Drinking Water Systems with Declining Populations: Lessons from Pennsylvania. Duke's institutional record establishes authorship and the 16-system/four-case scope: <https://scholars.duke.edu/publication/1565546>. Exact report full text remains unavailable through the checked publisher/repository routes; no claim is made that it lacks an equivalent regression or tariff appendix. The literature search is targeted, not exhaustive.

The October 2026 review inspected the Kentucky affordability report’s methods, rate-case examples, and ratemaking appendix in full text. That report already uses fixed-consumption charges and surcharges; this paper’s addition is the 2019–2024 linkage to accounts and a common reconstruction for the primary sample. Full-text sections of Allaire and Dinar (2022), Patterson and Doyle (2021), and Patterson, Bryson, and Doyle (2023) were accessible. For Cardoso and Wichman (2022), the author-hosted February 2022 working paper’s introduction and methods were inspected alongside the published bibliographic record. It is not treated as proof of identical wording in the published article.

Bash et al. (2020), Doyle et al. (2020), August et al. (2023), Mahmood and Lane (2026), and Ormsbee et al. (2026) were checked through accessible institutional, author-hosted, or publisher summaries and metadata where complete articles were unavailable. Search-index excerpts from the Pennsylvania PDF were also available, but do not constitute inspection of the complete report. `sources/cited_works.csv` records source-specific coverage and URLs.

The 40 endpoint tariff records were previously checked against original pages; this edition reuses that coding. Machine checks verify hashes and arithmetic, not human double coding. Research preparation, coding, source review, and document revision used AI assistance. The present review does not constitute independent human double coding or completed faculty review. `docs/evidence.md` describes documentary limits, and `docs/reproduction.md` reports execution and comparison results.

Mountain's refund installment and calendar-duration conflicts, Hyden's earlier phase-two first billing, and Powell's first surcharge invoice remain unresolved. These are within-period limits rather than new endpoint corrections. Seven account exclusions, category shares and physical territory continuity can affect the meaning or composition of the sample. The prior source memos distinguish those issues without imputed account reallocations or invoices.

7 Research files and execution

The analytical observations, eligibility rules, and specifications remain those of the v4 analysis. `README.md` identifies the files and execution commands; `docs/reproduction.md` gives the tested workflow and its scope. Current validation uses fixed references in `data/reference/` and requires no historical directory. Sources retain their identities and qualifications. Execution from compiled inputs does not reconstruct the original collection and screening process; availability of the source files does not establish redistribution permission.

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